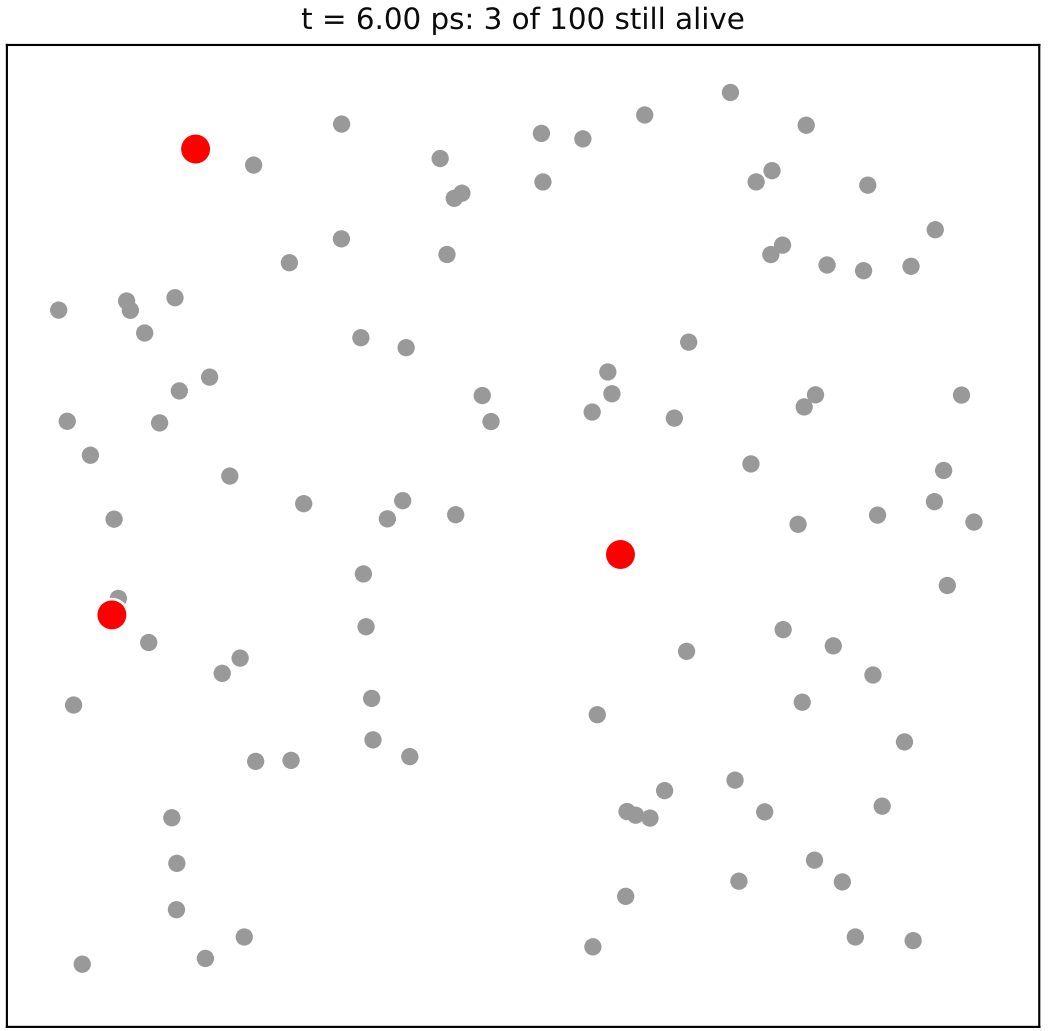
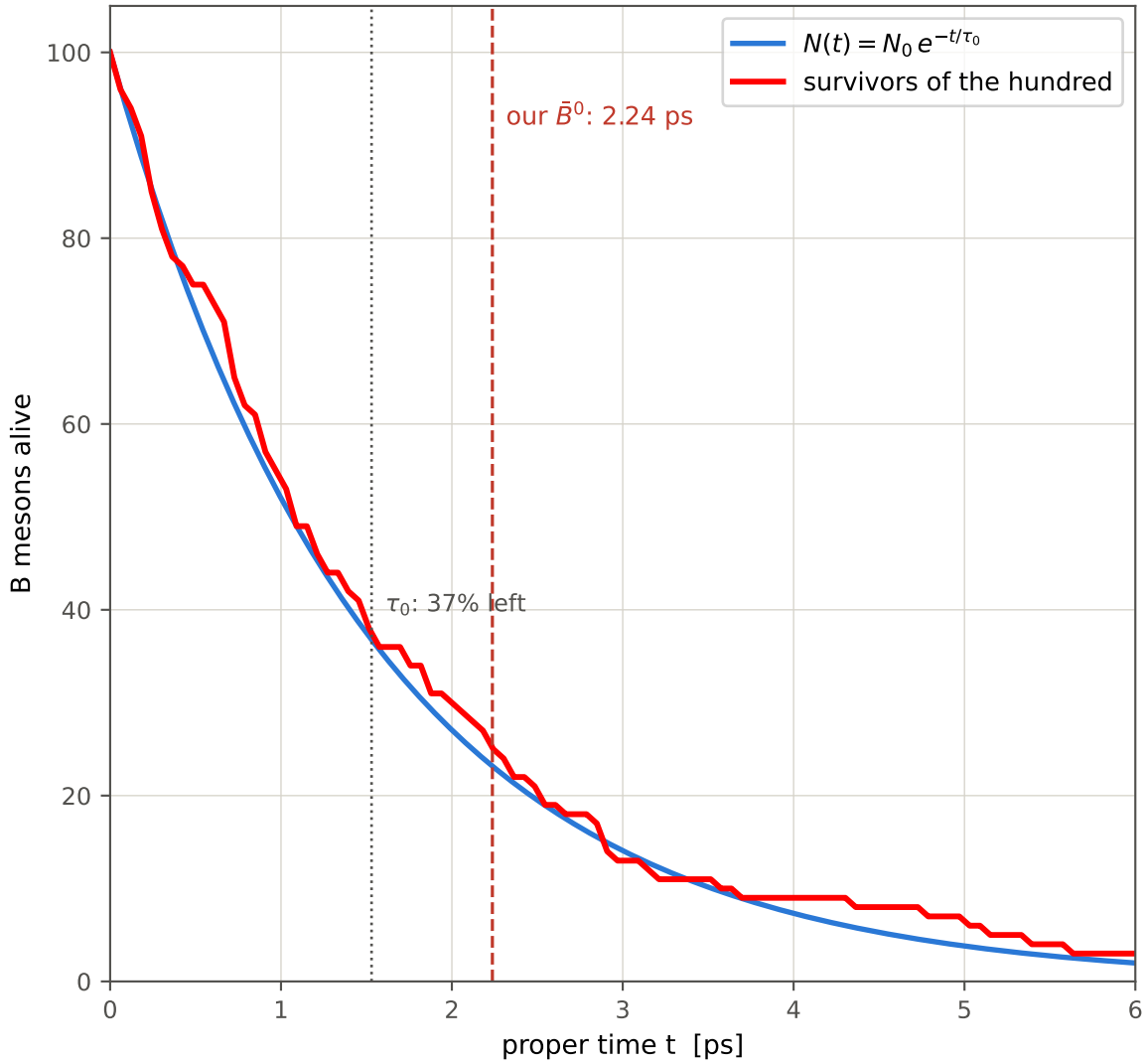
# Scene 1: the clock without memory

standalone_hadron_decays.py

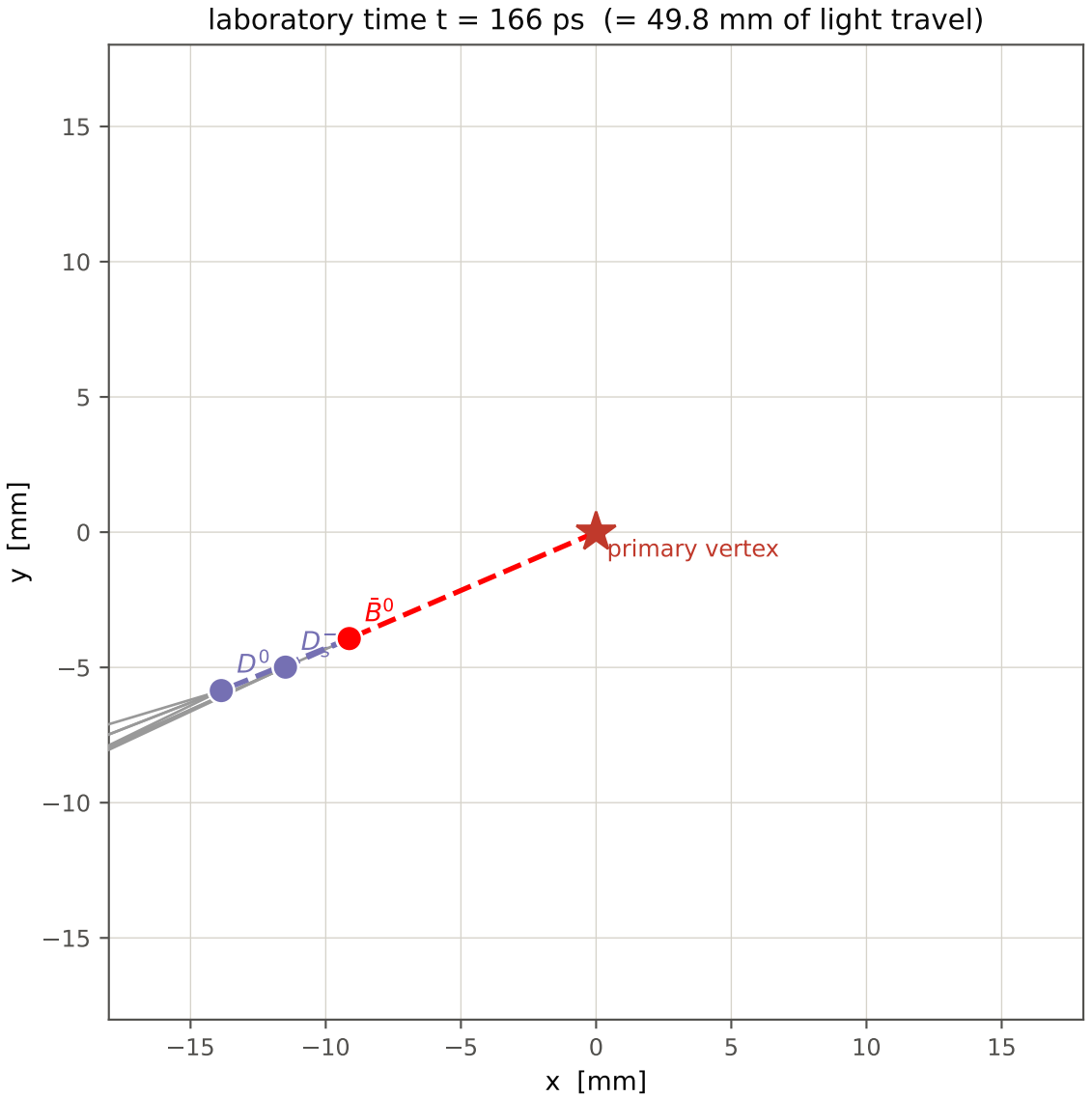


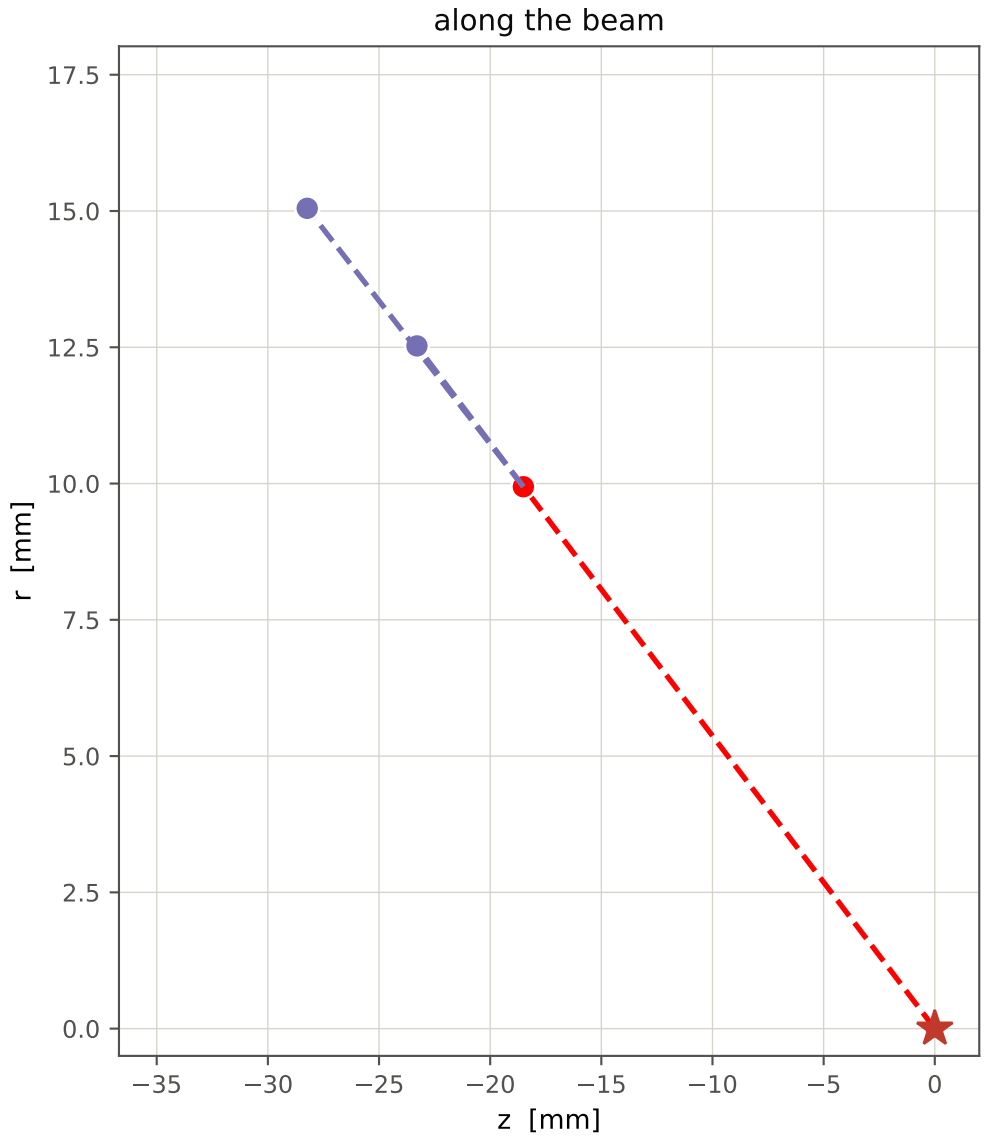




A hundred $B^0$ mesons born at t = 0: each decays at a random moment drawn from $e^{-t/\tau_0}$, $\tau_0$ = 1.53 ps. The survivors never age: the chance to decay in the next picosecond is the same at t = 0 and at t = 5 ps.
Our event's $\bar{B}^0$ drew t = 2.24 ps (1.46 lifetimes).

# Scene 2: the walk: a centimetre per picosecond

standalone_hadron_decays.py

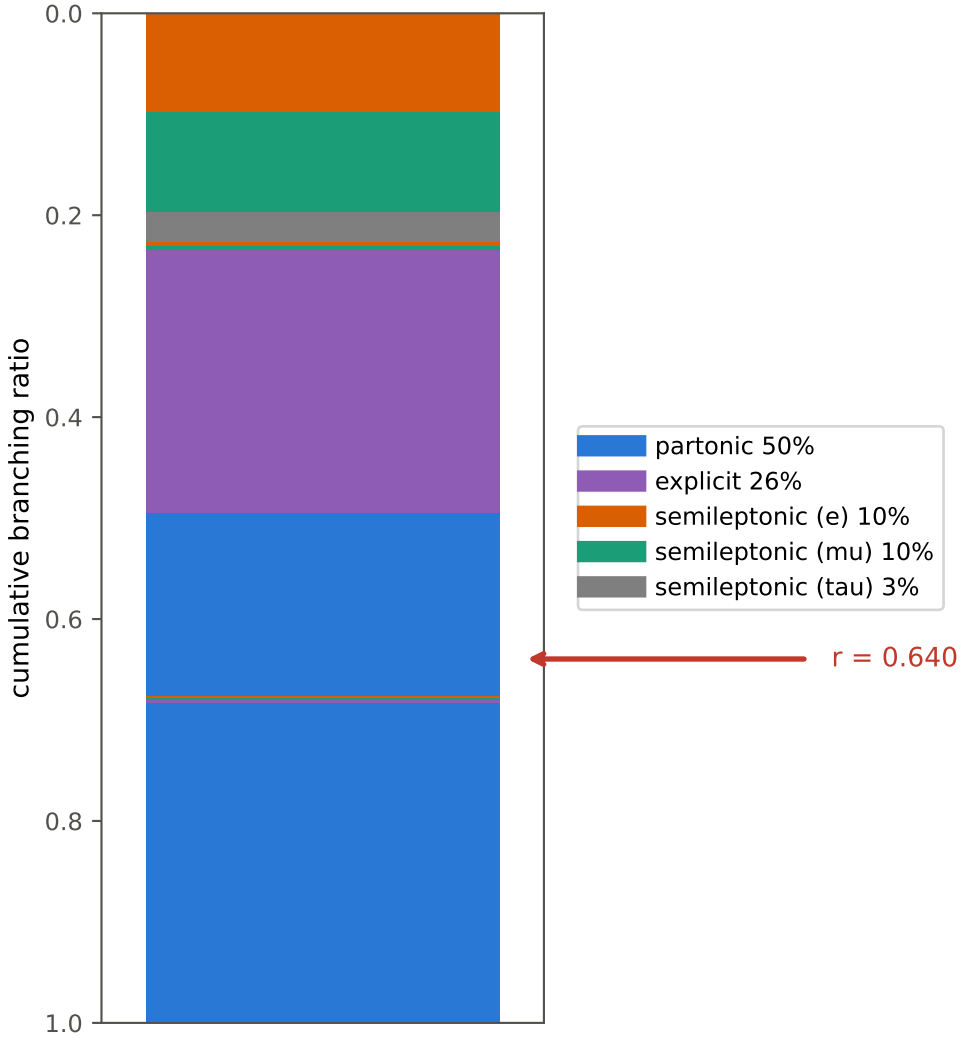



The $\bar{B}^0$ ($p_T$ = 78.3 GeV, $\gamma\beta$ = 31) leaves the primary vertex: its proper time t = 2.24 ps is stretched to 70 ps in the laboratory, a walk of L = 21.0 mm ($L_{xy}$ = 9.9 mm). It decays into D*_s- D*0 rho+; the D0 walks on another 11.0 mm.
The tracks of the 8 charged grandchildren start from the displaced vertices: 8 of them miss the beam line by more than 100 um.

# Scene 3: the decay table and the roll of the dice






the channel drawn (7.40%):

$\bar{B}^0 \to \bar{c}\, s\, c\, \bar{d}$

partons → hadrons (chapter 5's flavour machinery):

$D_s^{*-}\ D^{*0}\ \rho^+$

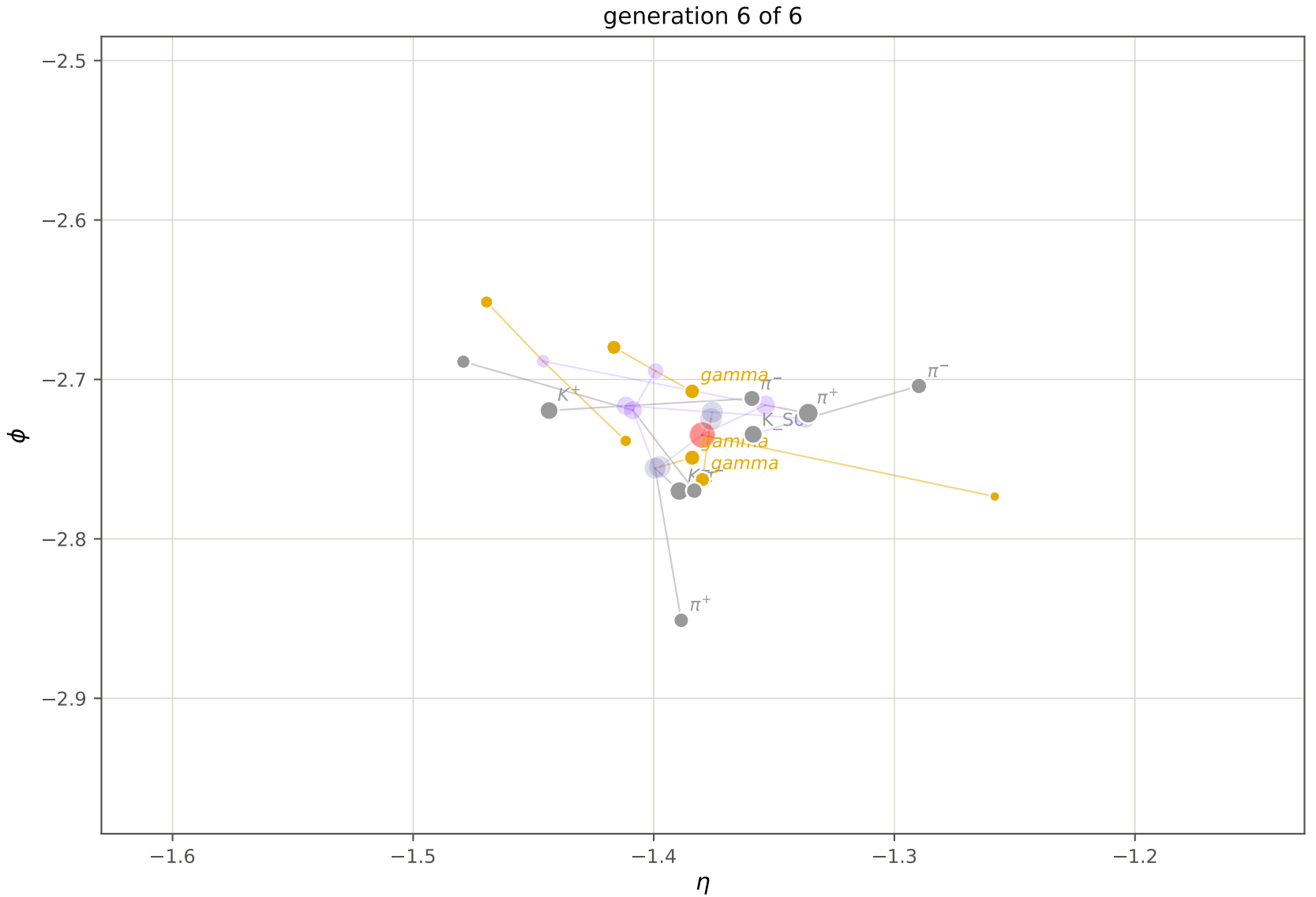
3 direct products, 15 final particles in the end

class: hadronic

The $\bar{B}^0$ has 870 open channels whose branching ratios add up to one. A random number r picks the channel in proportion to its branching ratio: ours lands on $\bar{B}^0 \to \bar{c}\, s\, c\, \bar{d}$ (7.4%, meMode 43).
The channel is written at quark level: the partons are paired into colour singlets and the string flavour machinery of chapter 5 turns them into hadrons: $D_s^{*-}\ D^{*0}\ \rho^+$.

# Scene 4: the cascade: grandchildren, great-grandchildren





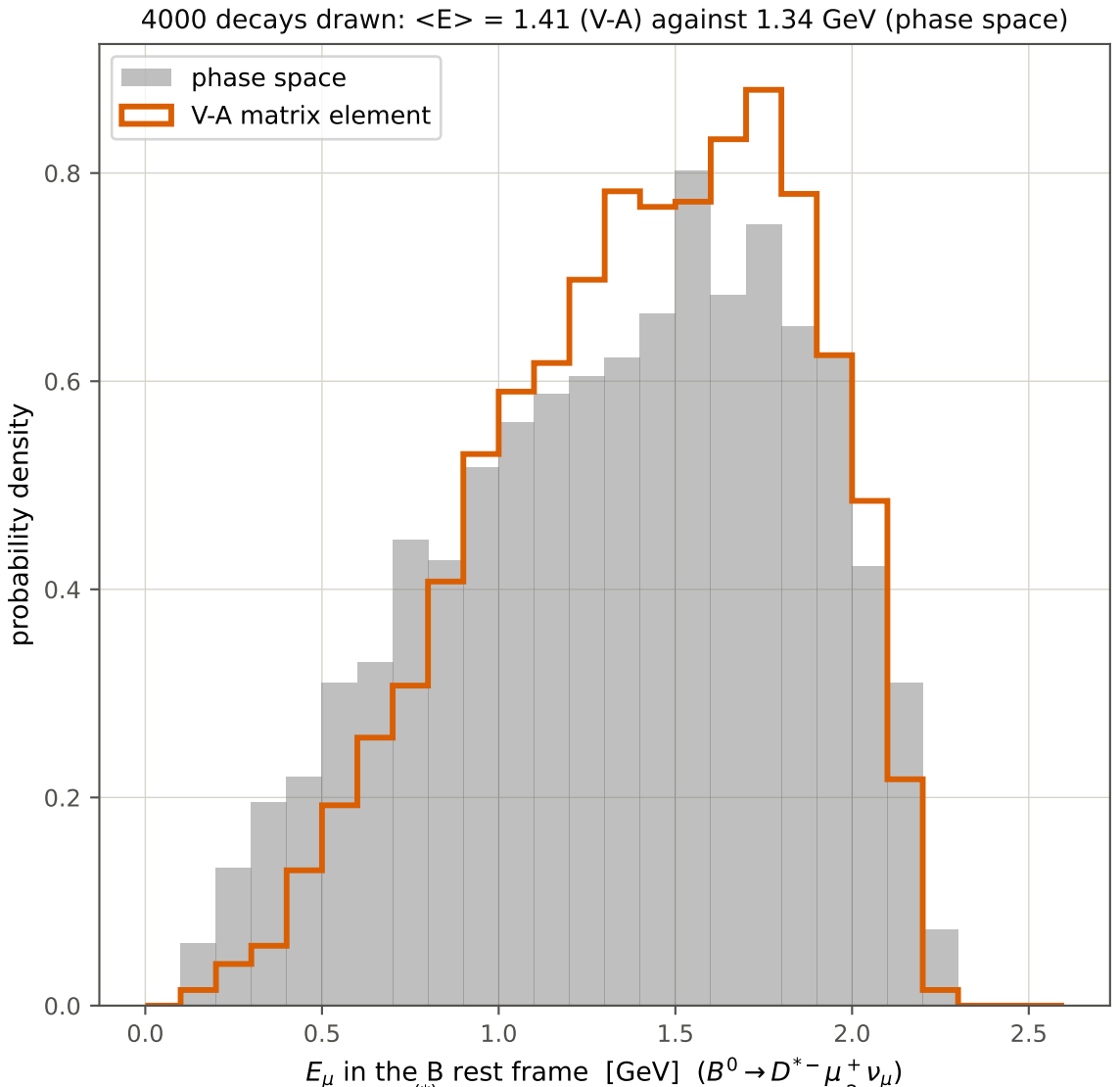

**particles alive: 16**

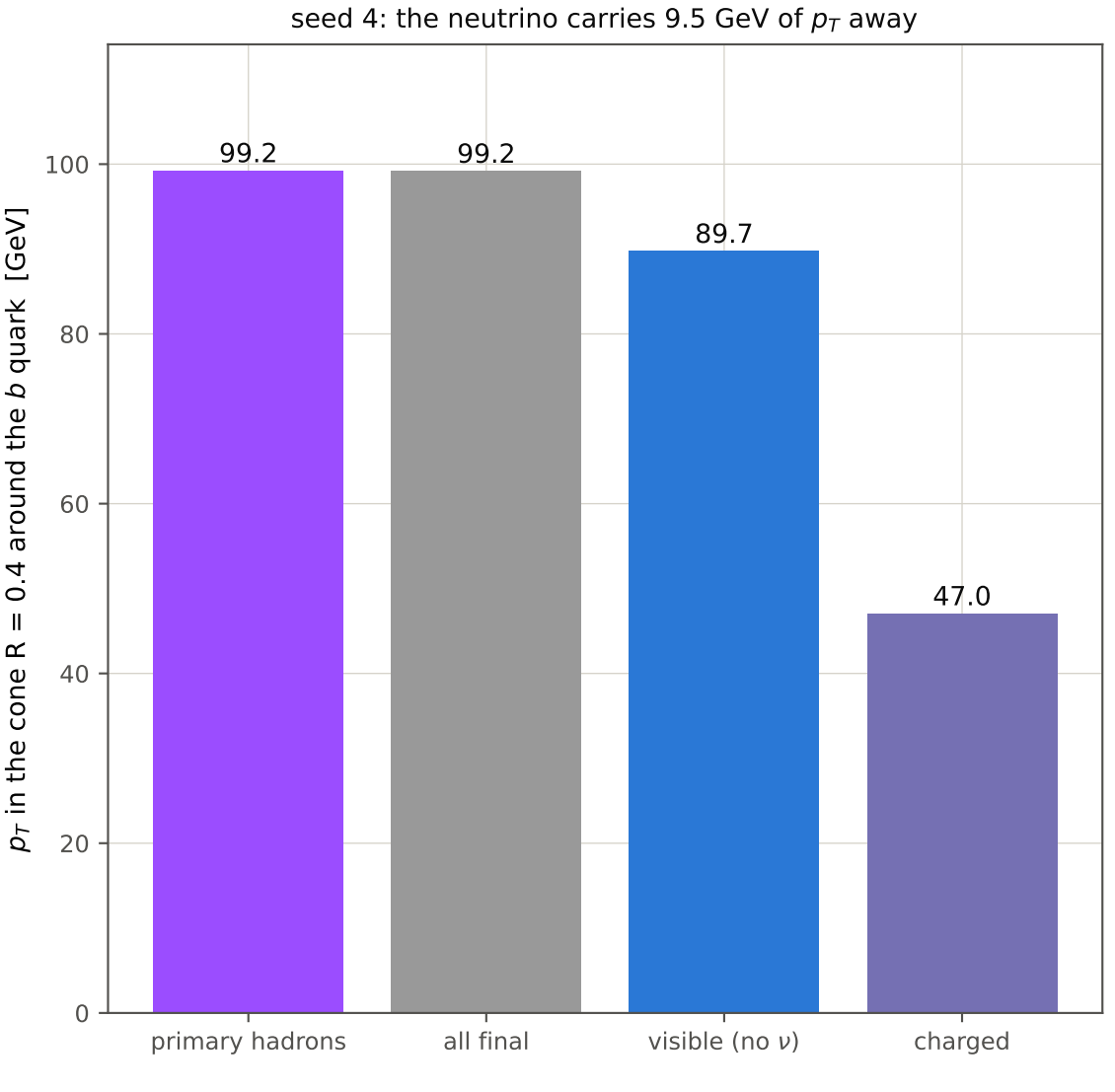
- *gamma* x 7
- $\pi^+$ x 3
- $\pi^-$ x 3
- $K^-$ x 1
- $K^+$ x 1
- K_S0 x 1

The $\bar{B}^{*0}$ of the string decays generation by generation: $\bar{B}^{*0} \to \bar{B}^0 \to D_s^{*-} \to D_s^-$ ... until only particles the detector can see are left: 15 of them (8 charged, 6 photons, 0 neutrinos), all within a few tenths in $\eta$ and $\phi$ of the original $b$ quark.

# Scene 5: the emigrant: a neutrino leaves the family

standalone_hadron_decays.py



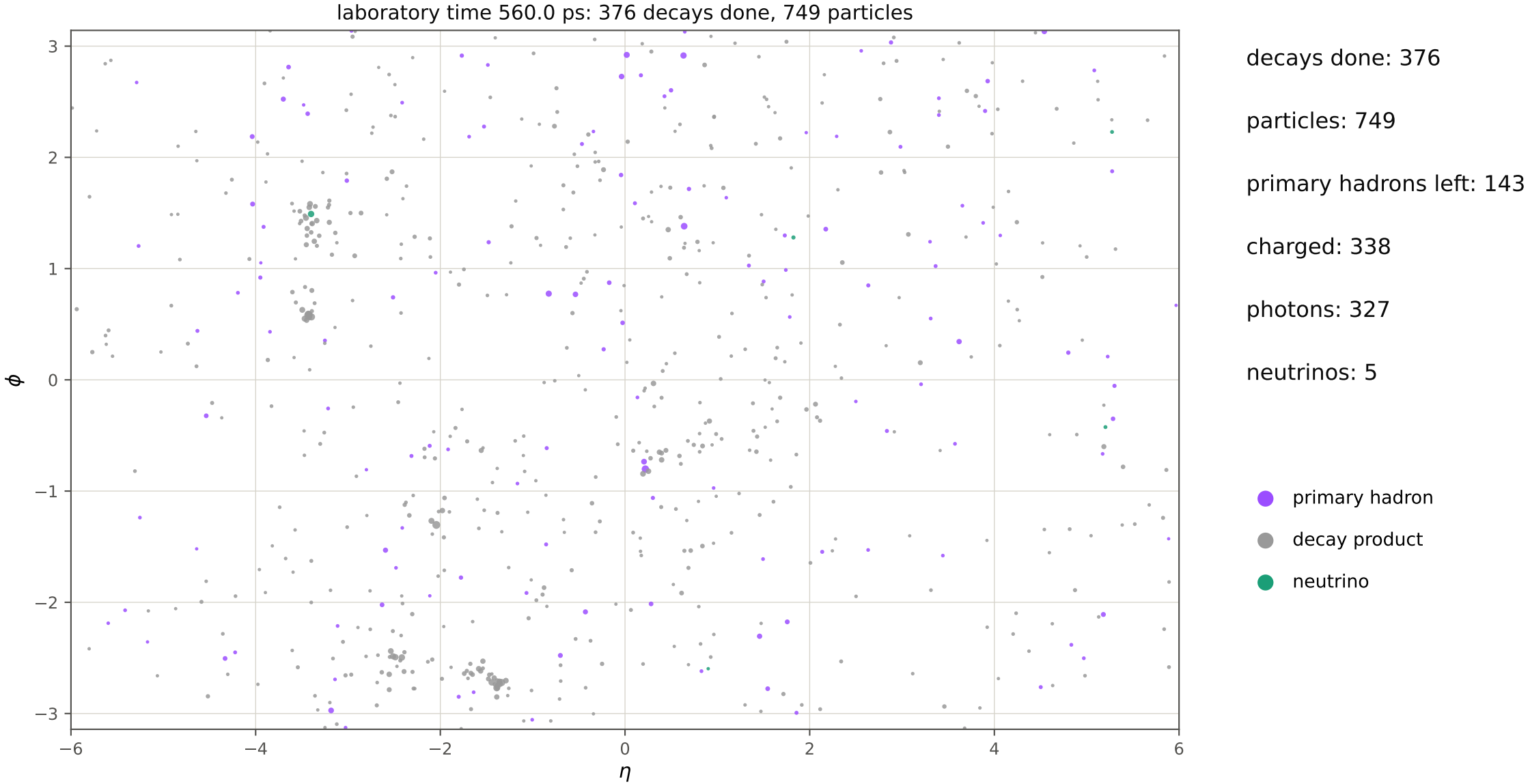




In a semileptonic decay $B \to D^{(*)} \ell \nu$ the V-A matrix element $|M|^2 \propto (p_B \cdot p_\nu)(p_D \cdot p_\ell)$ pushes the lepton and the neutrino to high energies. The neutrino then leaves the jet for good: the detector will measure less $p_T$ than the quark had.

Decaying the same hadrons with seed 4: $B^0 \to nu_{mu}\ mu^+\ D^{*-}$, the neutrino takes 19.8 GeV (12% of the B); the cone R = 0.4 drops from 99.2 to 89.7 GeV visible.

# Scene 6: the whole event: 365 hadrons become what the detector sees

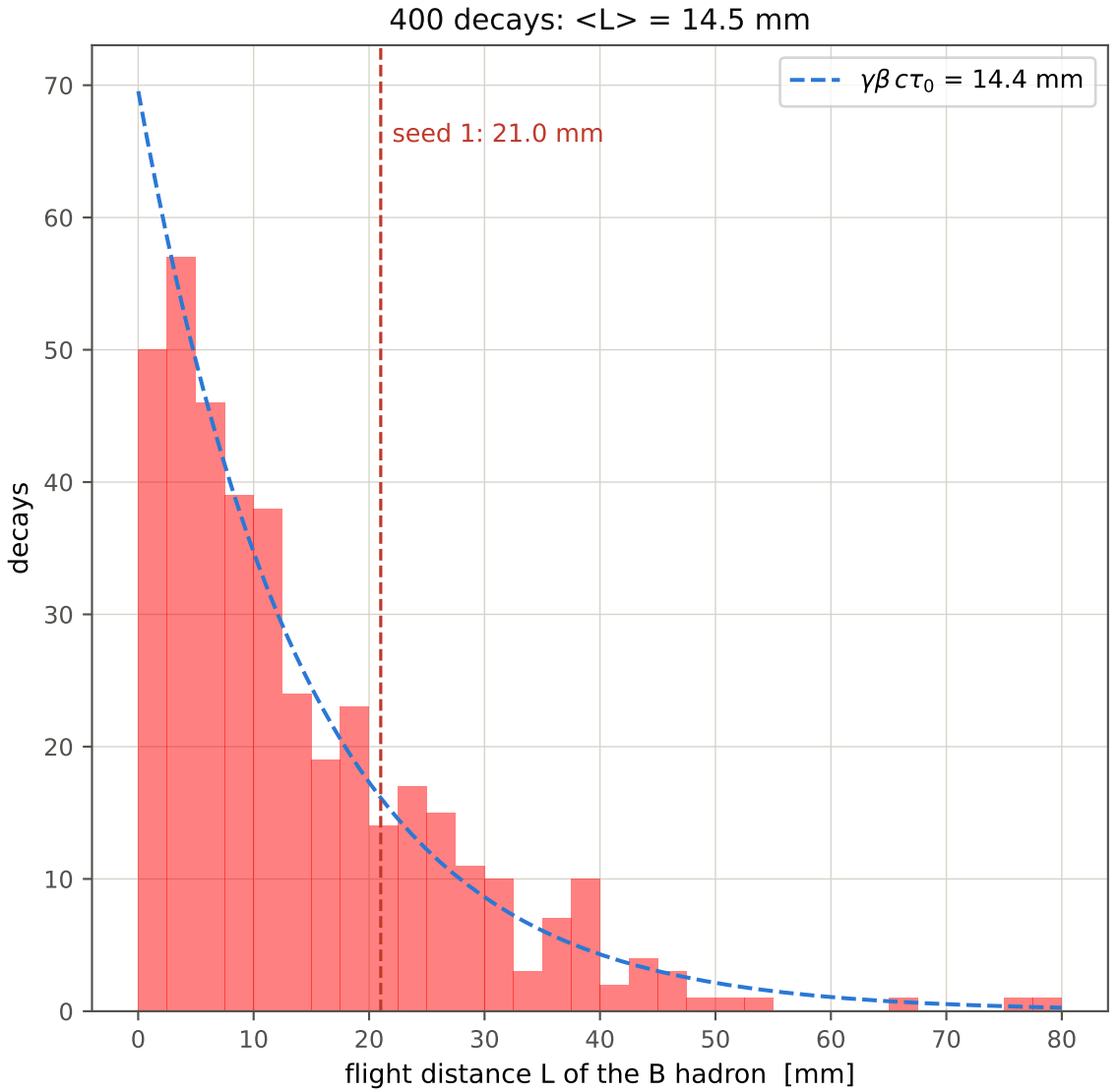
standalone_hadron_decays.py

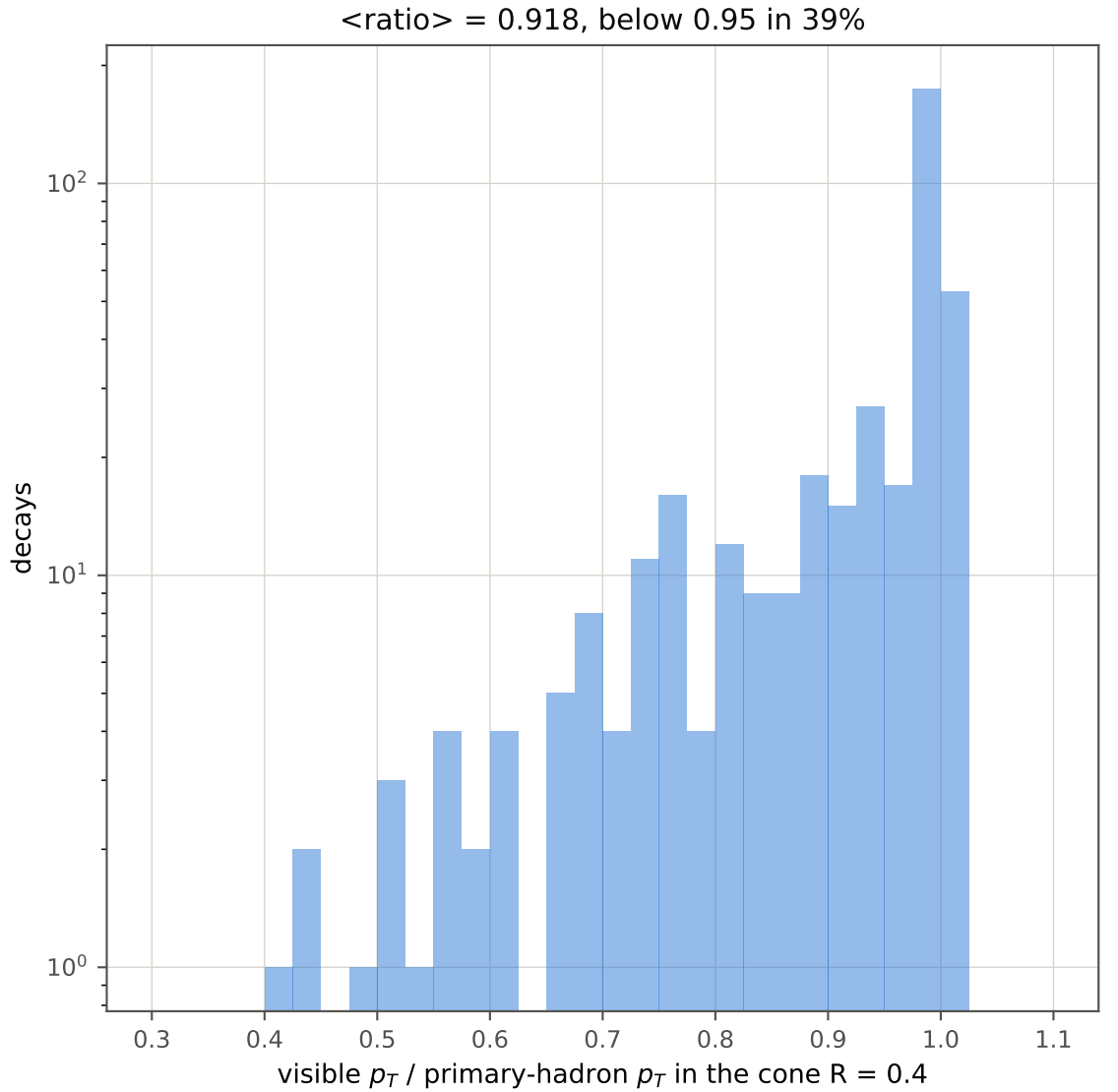



The decays in the order they happen in the laboratory: first the strong and electromagnetic ones at the primary vertex ($\rho$, $K^*$, $\pi^0$, ...), then the charm and bottom hadrons millimetres and centimetres away. 365 primary hadrons, 376 decays, 749 final particles: 338 charged, 327 photons, 5 neutrinos; 143 primaries survive unchanged.

# Scene 7: the same hadrons decayed again and again

standalone_hadron_decays.py




The 365 primary hadrons of the record decayed 400 times with different seeds. The flight distance of the *b* quark's B follows the exponential of scene 1 stretched by $\gamma\beta$ = 31; the visible $p_T$ of the cone R = 0.4 stays at the primary-hadron value in four decays out of five and drops by 10-50% when a neutrino leaves.